MODULE 36 · DOMAIN 3 - BUSINESS ENVIRONMENT

Risk I: plan, identify, analyse

Threats and opportunities, appetite and thresholds, finding and recording risks, qualitative analysis, EMV, decision trees and Monte Carlo.

DOMAIN III - Business Environment EXAM WEIGHT 26% MODULE 36 of 49

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A risk is uncertainty that matters. It can be good news too.

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Threat or opportunity?

THREAT

Uncertainty that could harm

- A supplier may deliver late
- Data may be poorer than expected
- A key person may leave
- Aim: reduce probability or impact

OPPORTUNITY

Uncertainty that could help

- A component may arrive early
- A new tool may speed testing
- A grant may become available
- Aim: increase probability or impact

THE TIEBREAKER

Identify opportunities as deliberately as threats. Both go in the risk register.

The risk management plan

- **Approach, roles and timing**: how risk is managed, by whom, and how often.
- **Categories**: often a risk breakdown structure, such as technical, external, organisational.
- **Definitions** of probability and impact, so everyone scores the same way.
- **Appetite and thresholds**, and how risks are escalated.
- **Funding** for risk responses and reserves, and how risk is reported.

> ON THE EXAM
>
> The outline's first risk enabler is to develop this plan. It makes risk management consistent.

Risk appetite or risk threshold?

RISK APPETITE

How much risk, in general

- The broad level of uncertainty the organisation will accept
- Set by leadership
- Qualitative: cautious, balanced, bold
- Shapes the whole approach

RISK THRESHOLD

The line in a specific measure

- A measurable limit on a specific objective
- 'Any risk over $100,000 impact'
- Triggers escalation or a response
- Turns appetite into action

THE TIEBREAKER

Appetite is the attitude; threshold is the number. A risk beyond a threshold is escalated, whatever its probability.

Identifying risks

TECHNIQUE	HOW IT HELPS
Brainstorming and workshops	Many perspectives, many risks, quickly
Interviews and expert judgement	Depth from people who have seen it before
Checklists and prompt lists	Categories so nothing obvious is missed
Assumption and constraint analysis	Every assumption is a potential risk
SWOT analysis	Strengths and weaknesses inside; opportunities and threats outside
Lessons learned and pre-mortem	What went wrong before; what could make this fail

ON THE EXAM

Identify risks with the team and stakeholders, early and throughout. Never alone at a desk.

Writing a clear risk statement

- **Cause**, **risk**, **effect**: 'Because X, Y may happen, which would lead to Z.'
- 'Because the old booking data has never been cleaned, migration errors may exceed five percent, delaying clinic go-live by up to three weeks.'
- 'Data problems' is not a risk statement. Nobody can score it or respond to it.
- A clear statement makes the probability, impact and response obvious.

> ON THE EXAM
>
> Cause, risk, effect. Vague risks get vague responses.

A risk register entry

FIELD	EXAMPLE: POOR IT SECURITY
Risk statement	Shared supplier admin passwords may let an attacker reach patient data
Category and owner	Security; IT security lead
Probability and impact	Medium; very high
Trigger	A shared account found in an access review
Response and status	Individual accounts, multi-factor login; in progress

ON THE EXAM

The outline names poor IT security as a register example. Each risk: an owner, a trigger, a response.

Qualitative analysis

- Score each risk's **probability** and **impact** using the plan's definitions.
- Combine them in a **probability and impact matrix** to rank risks.
- Consider **urgency** and **proximity**: how soon it could happen, and how soon you must act.
- Check data quality: a score based on a guess should be treated as uncertain itself.
- Quick, cheap, and done for every risk. It decides which few deserve more.

> ON THE EXAM
>
> Qualitative analysis first, for all risks. Quantitative analysis later, only for the few that justify it.

A probability and impact matrix

PROBABILITY	LOW IMPACT	MEDIUM IMPACT	HIGH IMPACT
High	Medium priority	High priority	High priority
Medium	Low priority	Medium priority	High priority
Low	Low priority	Low priority	Medium priority

ON THE EXAM

High probability and high impact first. But a low-probability, very high-impact risk still needs a response.

Expected monetary value

- **EMV = probability x impact**. Threats negative, opportunities positive.
- Threat: 30% x −$50,000 = **−$15,000**. Opportunity: 20% x +$40,000 = **+$8,000**.
- Together: −$15,000 + $8,000 = **−$7,000**.
- Summed across risks, EMV helps size the contingency reserve (module 26).

> ON THE EXAM
>
> Keep the signs: threats subtract, opportunities add. Most EMV errors are sign errors.

A decision tree: build or buy?

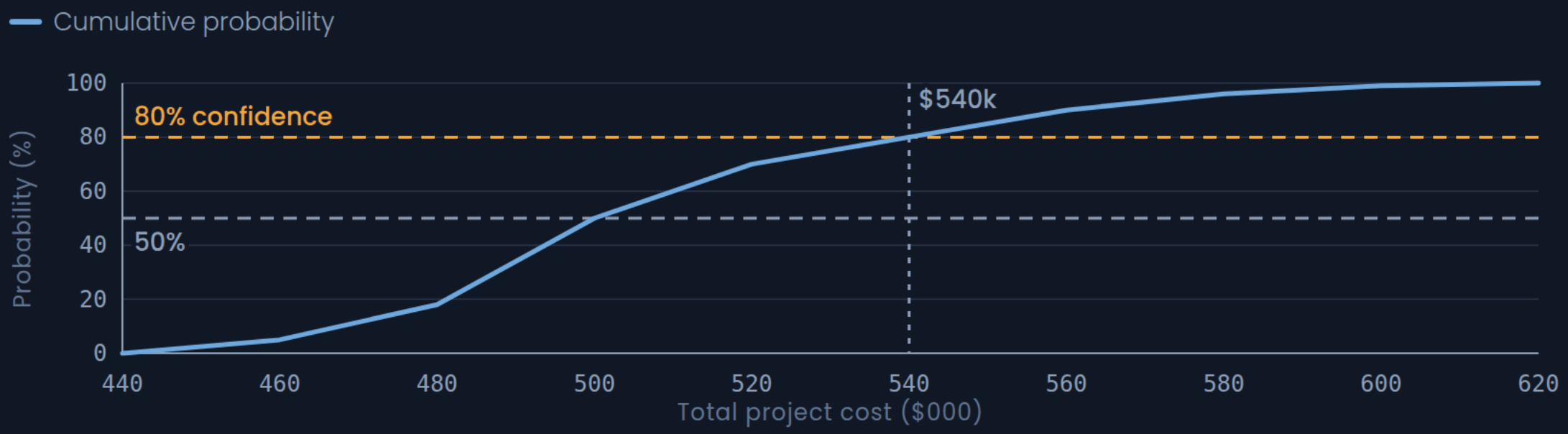
OPTION	COST	OUTCOMES	EMV
Build	-$200k	60% chance of $500k benefit; 40% chance of $300k	-200 + 300 + 120 = $220k
Buy	-$300k	90% chance of $450k benefit; 10% chance of $250k	-300 + 405 + 25 = $130k

ON THE EXAM

Compute EMV for each branch, including its cost. Choose the highest. Here: build, at $220k.

A Monte Carlo result

Probability of finishing at or below each cost



ON THE EXAM

Read across from the confidence level, then down to the cost. P80 here is $540,000.

Sensitivity analysis and quantitative choices

- **Sensitivity analysis** shows which risks move the outcome most, often as a **tornado diagram**.
- The widest bar at the top is the risk to manage first.
- Use quantitative analysis when the stakes, size or uncertainty justify the effort.
- Running simulations on every small project is effort without value.

> ON THE EXAM
>
> Qualitative for everything, quantitative for the few that matter, sensitivity to find the biggest drivers.

Risk on agile projects

- Short iterations and frequent reviews **reduce** risk by exposing problems early.
- A **risk-adjusted backlog** puts risk-reducing work, such as spikes, near the top.
- **Spikes**: timeboxed experiments to answer a risky question before committing.
- Risks are discussed in planning, daily meetings and retrospectives, not only in a register.

> ON THE EXAM
>
> Agile manages risk by learning fast. Risky, uncertain work goes early, not last.

Project Lantern: identifying and analysing risk

- A risk workshop with clinic managers, IT, the supplier and a patient representative.
- Forty-two risks identified, including three **opportunities**, such as a regional grant for digital health.
- Qualitative scoring for all; **Monte Carlo** on the migration and kiosk costs only.
- The contingency reserve was set at the P80 figure, and the steering committee agreed it.

> ON THE EXAM
>
> Wide identification, opportunities included, quantitative analysis only where it mattered.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Probabilities and impacts; total EMV?	Sum probability x impact; threats negative
A risk exceeds the plan's threshold	Escalate as the plan says, even if unlikely
Choose between options with uncertain outcomes	Decision tree: highest EMV, costs included
Many risks, little time	Qualitative first; quantitative for the top few
PM wrote the risk register alone	Identify with the team and stakeholders
Budget wanted at a confidence level	Read the Monte Carlo curve at that level

ON THE EXAM

Find risks widely, write them clearly, score them all, and quantify the few that matter.

> A project has three identified threats: a 20% chance of a $50,000 loss, a 10% chance of a $100,000 loss, and a 50% chance of a $10,000 loss.
>
> It also has one opportunity: a 25% chance of a $40,000 saving.

What is the project's total expected monetary value from these risks?

A −$25,000

B −$15,000

C −$35,000

D +$15,000

◆ PAUSE AND DECIDE

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What is the project's total expected monetary value from these risks?

A -$25,000

The threats only. The opportunity was left out.

B -$15,000

-10,000 - 10,000 - 5,000 + 10,000 = -15,000.

C -$35,000

Treats the opportunity as another loss. A sign error.

D +$15,000

The right size with the wrong sign.

ON THE EXAM

Threats negative, opportunities positive, then add. Every distractor is a sign or omission error.

> The risk management plan says any risk with a potential impact above $100,000 must be reported to the steering committee. The team identifies a risk with a 5% probability and a $150,000 impact.
>
> Its expected monetary value is only $7,500.

What should the project manager do?

A Accept the risk without reporting it, because its EMV is small.

B Leave it out of the register, because it is unlikely.

C Add $7,500 to contingency and take no further action.

D Record it and report it to the steering committee, because its impact exceeds the threshold.

◆ PAUSE AND DECIDE

What should the project manager do?

A Accept quietly: small EMV.

The threshold is on impact, not EMV. The plan requires reporting.

B Leave it out.

Unlikely risks still belong in the register.

C Add contingency only.

May be part of the response, but skips the required escalation.

D Record and report per the threshold.

Follows the risk management plan, whatever the probability.

THE SIGNAL

A threshold defined on impact is crossed by impact alone. Follow the plan and escalate.

> A team must choose between building a component or buying one. Building costs $200,000, with a 60% chance of $500,000 benefit and a 40% chance of $300,000.
>
> Buying costs $300,000, with a 90% chance of $450,000 benefit and a 10% chance of $250,000.

Based on expected monetary value, which option should be chosen?

A. Build, with an EMV of $220,000.

B. Buy, with an EMV of $130,000.

C. Buy, because it has the higher chance of success.

D. Build, because it has the lower upfront cost.

◆ PAUSE AND DECIDE

Based on expected monetary value, which option should be chosen?

A Build, EMV $220,000.
-200 + (0.6 x 500) + (0.4 x 300) = 220. The higher EMV.

B Buy, EMV $130,000.
Correct figure for buy, but lower than build.

C Buy: higher success chance.
Probability alone ignores both cost and the size of benefits.

D Build: lower upfront cost.
Right choice, wrong reason. Cost alone ignores benefits.

THE SIGNAL

Compute each branch's EMV including its cost, and pick the highest. The question said 'based on EMV'.

> A risk workshop has identified sixty risks for a large infrastructure project. The steering committee meets in a week.
>
> The team has a Monte Carlo tool but limited time to prepare data.

What should the project manager do next?

A Run a quantitative analysis on all sixty risks.

B Plan detailed responses for all sixty risks straight away.

C Score all sixty qualitatively to prioritise them, then analyse the top few quantitatively.

D Focus only on the risks with the highest impact, whatever their probability.

◆ PAUSE AND DECIDE

What should the project manager do next?

A Quantify all sixty.
Far too much effort; most risks do not justify it.

B Respond to all sixty.
Skips analysis; effort goes equally to trivial and serious risks.

C Qualitative for all; quantitative for the top few.
The standard sequence, and it fits the week available.

D Highest impact only.
Ignores probability, the other half of the score.

THE SIGNAL

Qualitative analysis first, for everything. Then quantify only what matters most.

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> A project manager has written the risk register alone over a weekend, using a template from a previous project.
>
> The team and stakeholders have not yet been involved. The register is due to go to the sponsor.

What should the project manager do before submitting it?

A Submit it, since it follows the organisation's template.

B Hold a risk identification workshop with the team and key stakeholders, and update the register.

C Copy in the previous project's risks as well, to be thorough.

D Wait until issues arise, then add them to the register.

◆ PAUSE AND DECIDE

What should the project manager do before submitting it?

A Submit it.

One person's view misses the risks others can see.

B Workshop with team and stakeholders.

Identification is a group activity. It finds far more, and builds ownership.

C Copy old risks.

Useful input, but no substitute for this project's people.

D Wait for issues.

Turns risk management into issue management.

THE SIGNAL

Risks are identified with the team and stakeholders. A register written alone is half a register.

> A Monte Carlo simulation of a project's cost shows a 50% probability of finishing at or below $500,000, an 80% probability at or below $540,000, and a 100% probability at or below $600,000.
>
> The sponsor wants a budget with 80% confidence.

What budget should the project manager recommend?

A $500,000

B $600,000

C $480,000

D $540,000

◆ PAUSE AND DECIDE

What budget should the project manager recommend?

A $500,000

Only 50% confidence: a coin toss.

B $600,000

Near-certainty. More reserve than the sponsor asked for.

C $480,000

Below the median; well under 50% confidence.

D $540,000

The P80 figure: 80% chance of finishing at or below it.

THE SIGNAL

Read the simulation at the confidence level asked for. P80 here is $540,000.

Find risks widely. Write them clearly. **Score them all**, and quantify the few that matter.

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